When two or more sets combine together to form one set under the given conditions, then operations on sets can be carried out. We can visualize the relationship between sets and set operations using Venn diagram.

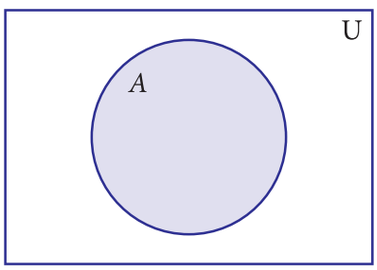
John Venn was an English mathematician. He invented Venn diagrams which pictorially represent the relations between sets.Venn diagrams are used in the field of Set Theory, Probability, Statistics, Logic and Computer Science.

## 1.5.1 Complement of a Set

The Complement of a set $A$ is the set of all elements of U (the universal set) that are not in $A$.

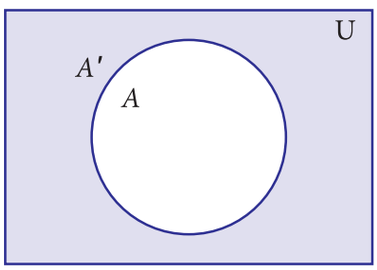
It is denoted by $A'$ or $A^c$. In symbols $A' = \{x : x \in U, x \notin A\}$

### Venn diagram for complement of a set




$A$ (shaded region)

Fig. 1.4




$A'$ (shaded region)

Fig. 1.5

For example,

If U = {all boys in a class} and $A$= {boys who play Cricket}, then complement of the set $A$ is $A'$= {boys who do not play Cricket}.

**Example 1.10**

If U = $\{c, d, e, f, g, h, i, j\}$ and $A = \{c, d, g, j\}$ , find $A'$.

***Solution***

U = $\{c, d, e, f, g, h, i, j\}$, $A = \{c, d, g, j\}$

$A' = \{e, f, h, i\}$

**Note**

- $(A')' = A$
- $U' = \varnothing$
- $\varnothing' = U$

## 1.5.2 Union of Two Sets

The union of two sets $A$ and B is the set of all elements which are either in $A$ or in $B$ or in both. It is denoted by $A \cup B$ and read as $A$ union $B$.

In symbol, $A \cup B = \{x : x \in A \text{ or } x \in B\}$



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